Project Review: tanerjn/mp

GitHub repository: https://github.com/tanerjn/mp | Branch reviewed: main | Review date: October 22, 2026

1. Project overview

The repository is focused on spacecraft maneuver planning. Its directory and document names suggest CubeSat collision-avoidance research, with low-Earth-orbit (LEO) examples and a second phase involving geostationary-orbit (GEO) scenarios. The project appears to explore safe separation planning subject to propulsion constraints. These are preliminary interpretations based on available filenames and one readable configuration file, not a verified account of the complete algorithms.

2. Repository organization

README.md - a brief description: "mp" and "maneuver planning".
Maneuver_Planning/config.py - readable orbital and maneuver parameters.
__Maneuver_Planning/ - documentation, planners, validation-related files, results, and Phase_2_Maneuver_Planning.
documentation/ - CubeSat project handoff documents, future GEO scenario notes, and dated session reports.
planners/ - filenames for full-order, reduced-order, and hybrid planners.
Phase_2_Maneuver_Planning/ - GEO configuration, dynamics, encounters, examples, validation, tests, and results.
results/ - LEO baseline filenames and GEO runs labeled 6h, 12h, and 24h.

3. Confirmed LEO example configuration

Parameter	Value
Orbit altitude	500 km
Earth gravitational parameter	3.986004418 × 10^14 m³/s²
Closest approach time	1,200 s
Burn time	300 s
Simulation end	1,800 s
Required separation	100 m
Planning separation	105 m
Maximum delta-v	0.5 m/s

These values come directly from Maneuver_Planning/config.py. The 105 m planning separation is 5 m above the required minimum.

4. Development areas indicated by the files

Area	Evidence in repository
LEO maneuver planning	500 km baseline configuration and result filenames
Planner approaches	full_order.py, reduced_order.py, hybrid.py filenames
GEO scenario development	config_geo.py and GEO scenario document filenames
Simulation horizons	GEO result filenames labeled 6h, 12h, and 24h
Validation	validation.py and tests directory names
Documentation	Handoff documents and dated session reports

5. Repository limitations

Many source and documentation paths end in _Error.txt. When opened, sampled files contained a download failure message reporting TooManyRequestsMeTAException, not the intended source code or documentation. As a result, the detailed planning algorithms, experiment results, validation outcomes, and project milestones could not be independently confirmed. The root README is minimal.

6. Assessment and recommended next step

The repository shows a plausible progression from a 500 km LEO demonstration to GEO-oriented maneuver-planning studies, with several planner variants and recorded experiment filenames. The most important next step is to obtain readable versions of the original Python modules and handoff notes, then trace the system architecture, objective functions, constraints, simulations, and validation tests. No repository files were modified during this review.

Scope note: The mp.zip archive was intentionally excluded from review at the user's request because it duplicates the repository contents.